

Dataset proposal for pre-validation

Course: Interpretability, Stability, and Algorithmic Fairness Dataset: NIJ 2021 Recidivism Forecasting Challenge, Georgia parole cohort Source: <https://nij.ojp.gov/funding/recidivism-forecasting-challenge>

Proposed client question

At the start of community supervision, which people are most likely to experience a new arrest within three years? A hypothetical community-supervision software vendor would evaluate the estimate for prioritizing voluntary, beneficial support such as employment, housing, or treatment services. It would not be used to increase surveillance, restrictions, detention, or punishment.

Why the dataset fits the project

The post-challenge file contains 25,835 records, a binary three-year target, an official 70/30 train/test flag, and 32 baseline person-level variables. It supports the required comparison of:

1. Logistic regression as the white-box model;
2. XGBoost as the conventional machine-learning model;
3. TabICLv2 as the open tabular foundation model.

It also contains race and gender, enabling explicit subgroup audits. Those attributes will be used only for evaluation, not as model inputs. Residence geography will also be excluded because it can proxy protected status.

Evaluation design

Predictive performance includes probability baselines, ROC AUC, average precision, Brier score, log loss, calibration error, paired bootstrap differences and training-only fixed-configuration CV. Economic value is a scenario based on assumed capacity, costs and effectiveness. Interpretability includes SHAP, LIME, XPER, a surrogate and PDP/ICE. Stability compares bootstrap refits; fairness covers race/gender and descriptive intersections. The evaluation partition was repeatedly inspected during development, so independent validation remains necessary.

The cumulative three-year target is a course-project adaptation. NIJ evaluated annual conditional forecasts on changing cohorts; our performance is not comparable to its leaderboard.

Key validity guardrail

Variables describing violations, drug tests, employment, programs, and residence changes accrue after supervision begins. They will be excluded from the prediction because using them for a baseline score would leak post-decision information. The outcome is arrest rather than offending, so the report will discuss construct bias, exposure to policing, historical context, and limited transportability beyond Georgia releases from 2013–2015.